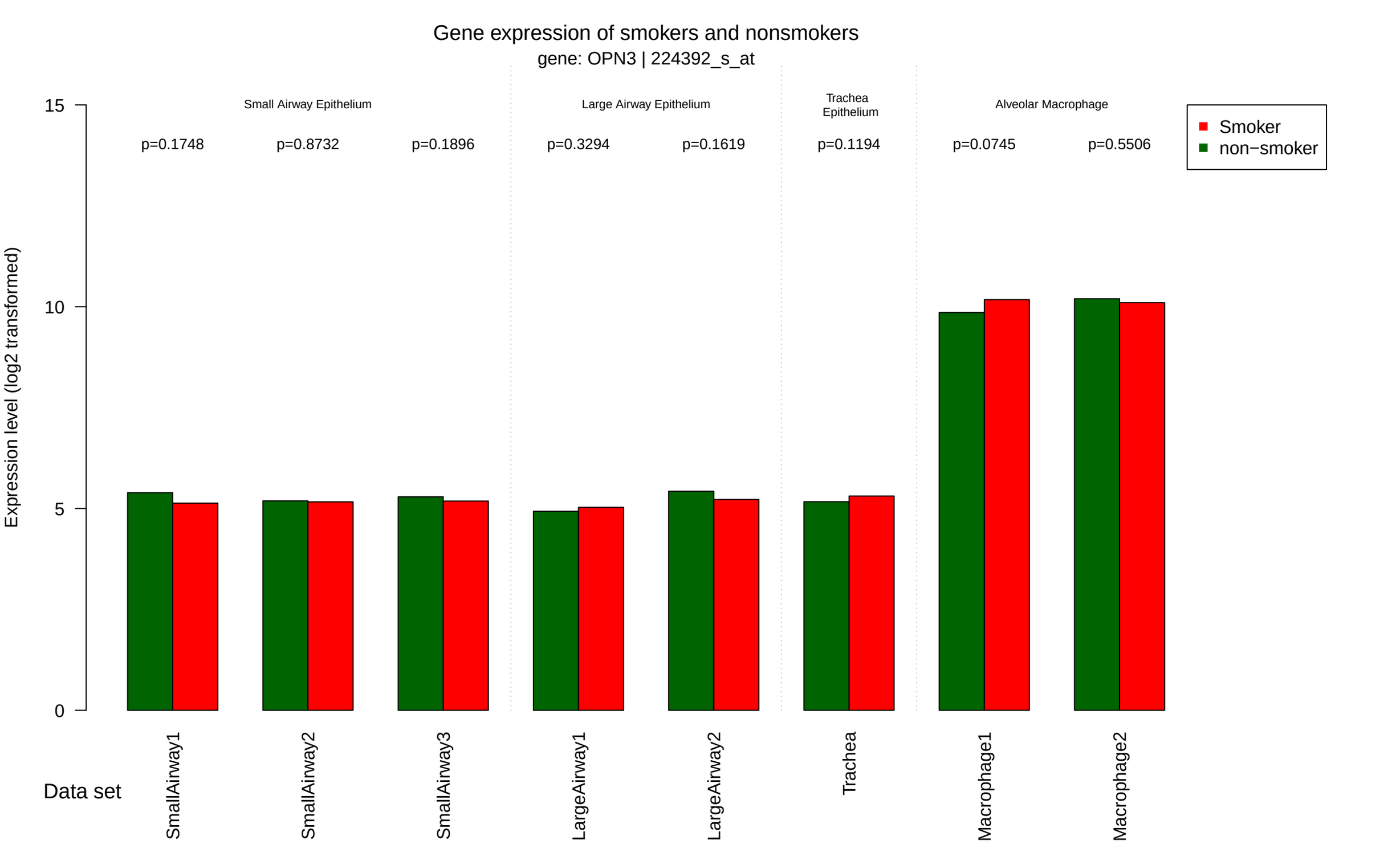
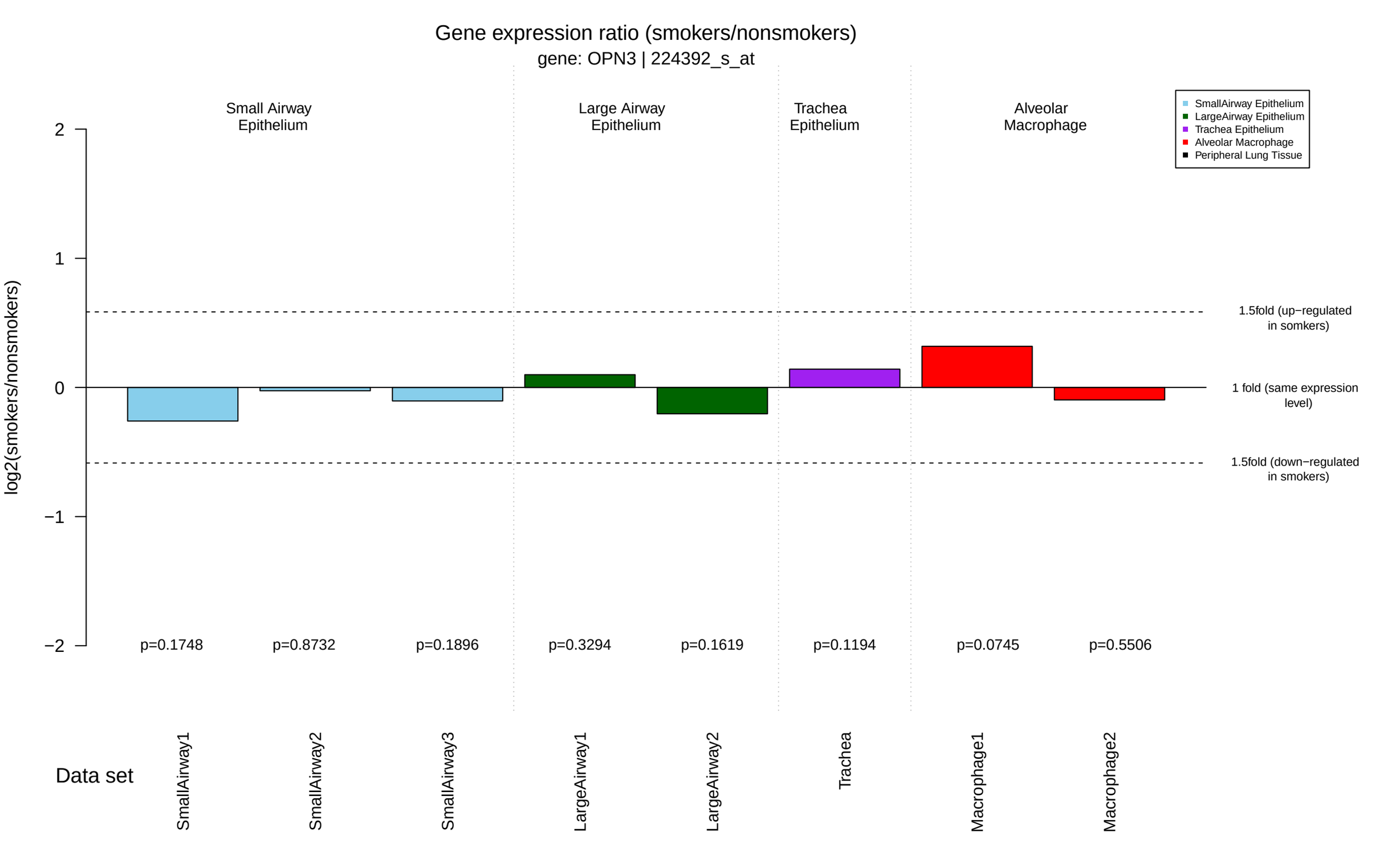
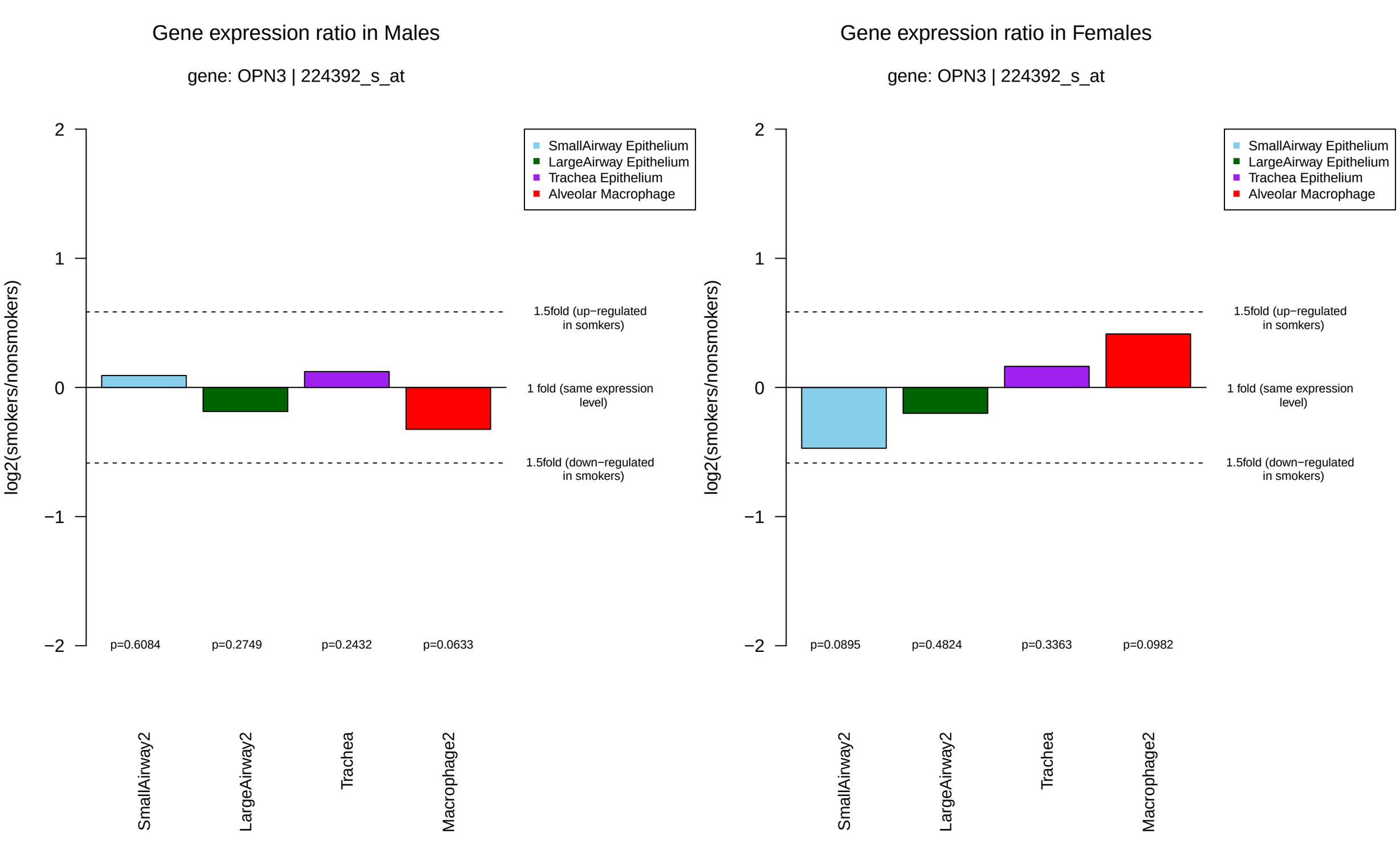
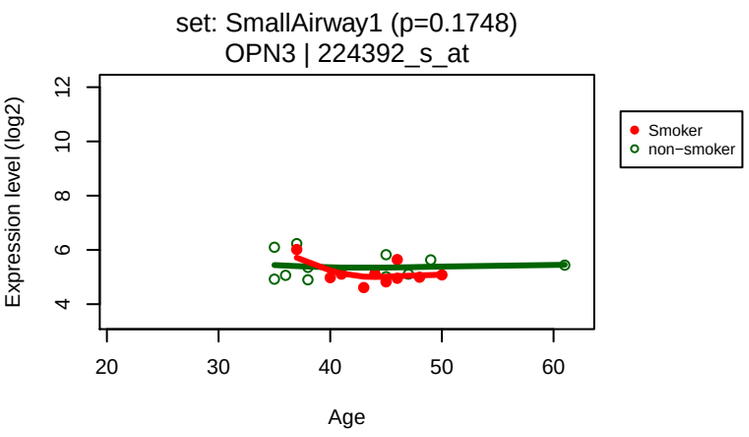
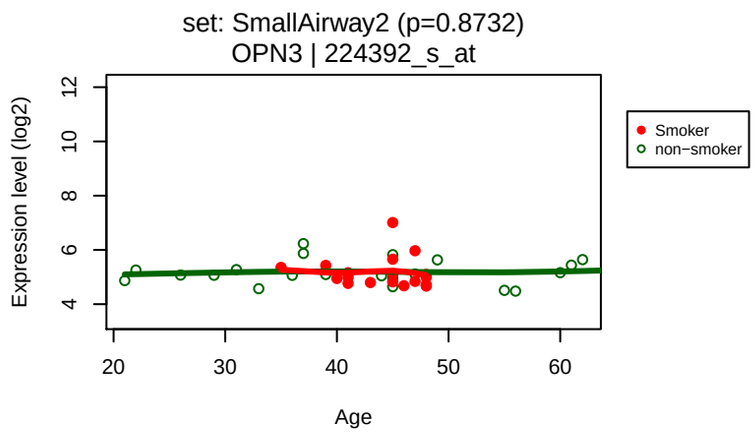
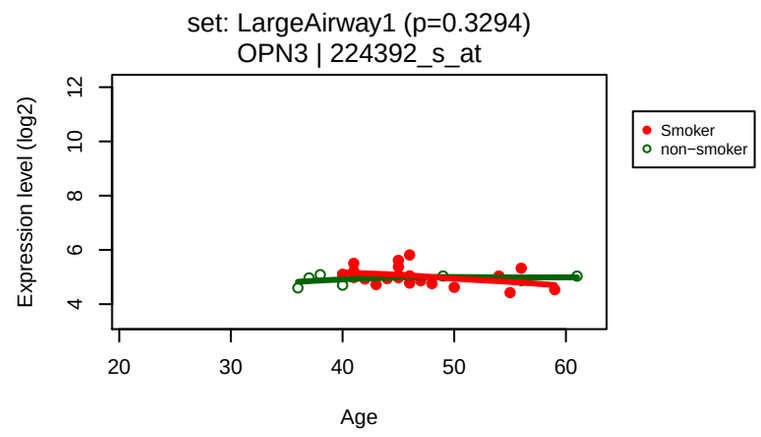

Gene expression of smokers and nonsmokers
gene: OPN3 | 224392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1748
p=0.8732
p=0.1896
p=0.3294
p=0.1619
p=0.1194
p=0.0745
p=0.5506
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

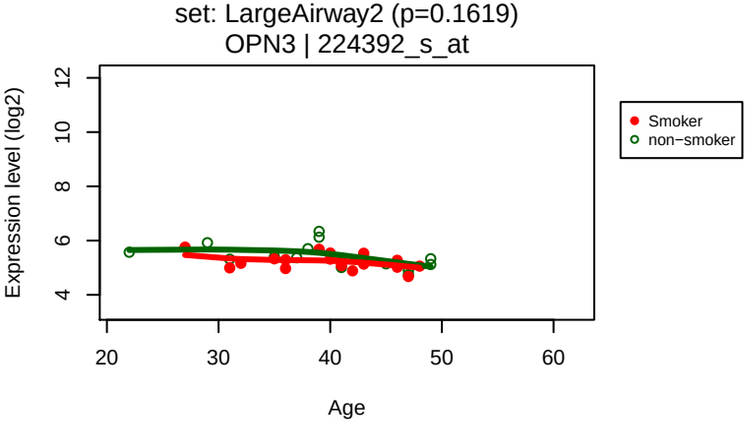

Gene expression ratio (smokers/nonsmokers)
gene: OPN3 | 224392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1748
p=0.8732
p=0.1896
p=0.3294
p=0.1619
p=0.1194
p=0.0745
p=0.5506
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

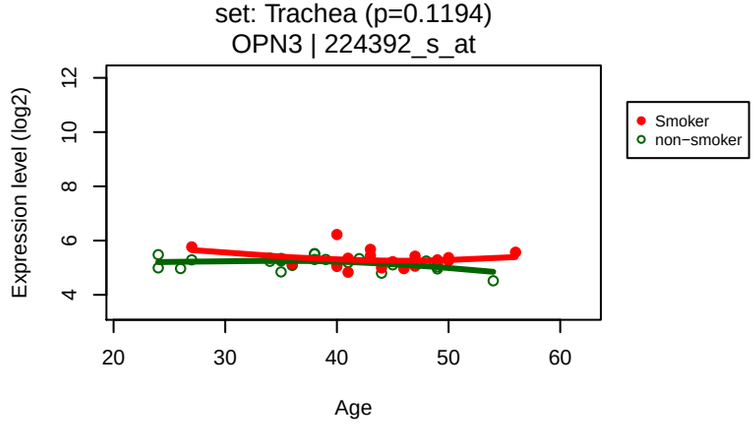

Gene expression ratio in Males
gene: OPN3 | 224392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6084
p=0.2749
p=0.2432
p=0.0633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPN3 | 224392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0895
p=0.4824
p=0.3363
p=0.0982
SmallAirway2
LargeAirway2
Trachea
Macrophage2

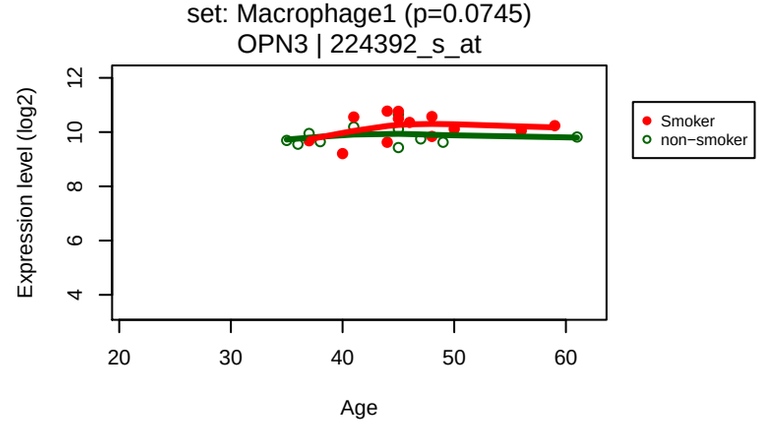

set: SmallAirway1 (p=0.1748)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker

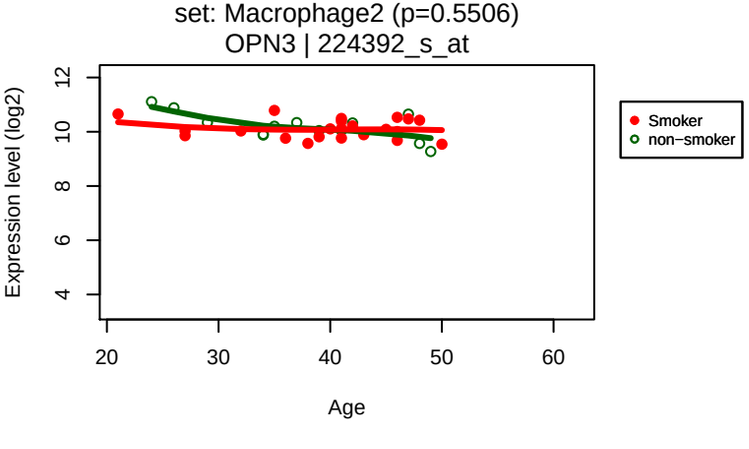

set: SmallAirway2 (p=0.8732)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker

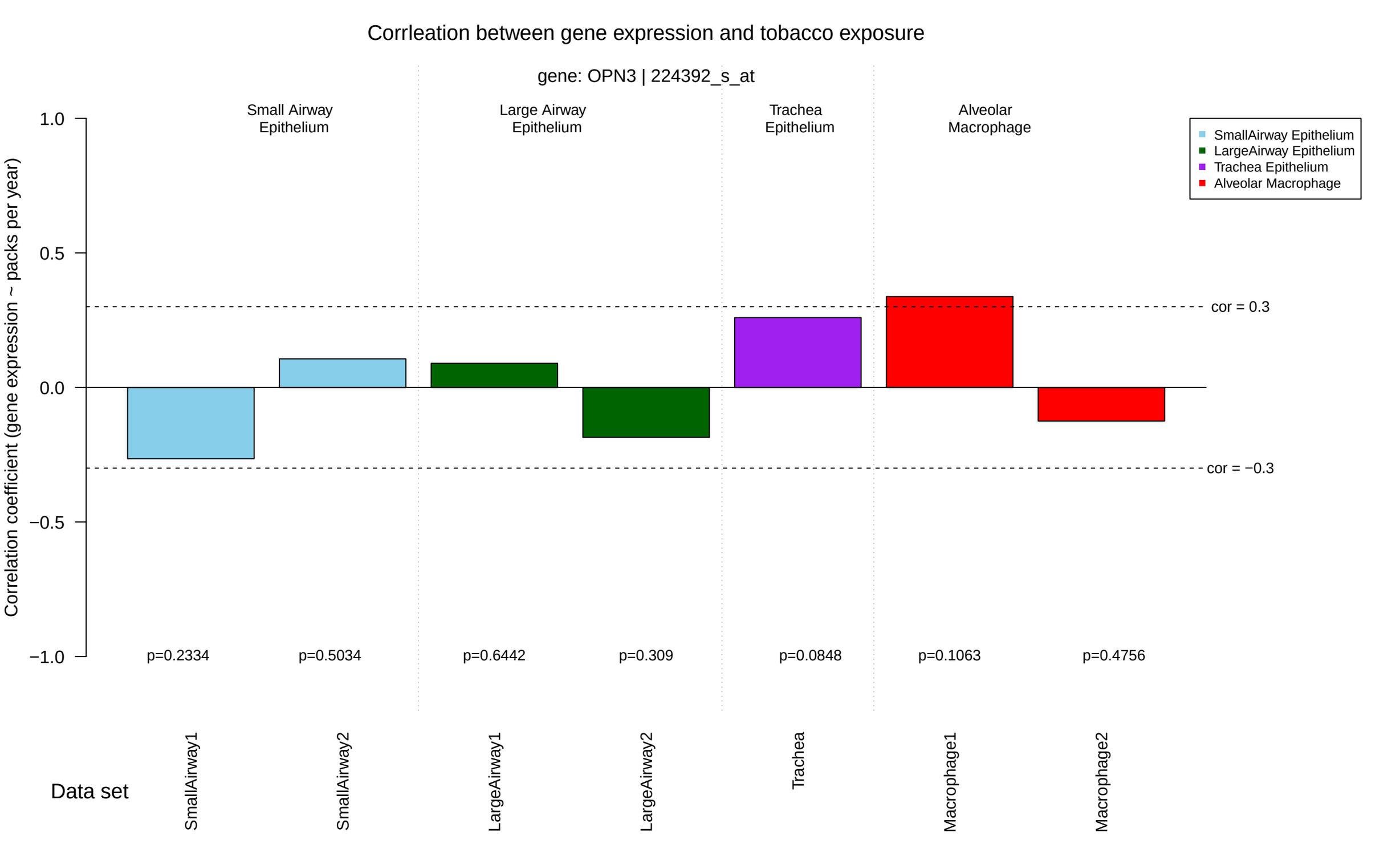

set: LargeAirway1 (p=0.3294)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1619)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1194)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0745)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5506)
OPN3 | 224392_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPN3 | 224392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2334
p=0.5034
p=0.6442
p=0.309
p=0.0848
p=0.1063
p=0.4756
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2